

FORESIGHT

Data-Quality & EDA Insight Memo

Demand & Inventory Intelligence | NorthBay Living

Prepared from the executed project notebook and processed sales outputs.

Purpose

Give the planning team a clear view of what was found in the sales data before forecasting and risk scoring were applied. The memo focuses on data reliability, demand behaviour, product concentration and slow-moving inventory.

1. Data quality review

The analysis started with a 100,000-row transaction sample containing 11 fields. The first pass was deliberately conservative: check the structure, validate numeric fields, inspect dates and identify duplicates before changing anything.

Check	Finding	Treatment
Rows / columns	100,000 / 11	Retained for analysis sample
Duplicate rows	137 (0.14%)	Removed
Missing promo_id	79,308 rows (79.31%)	Filled as NO_PROMO
Invalid dates	0	No removal required
Invalid quantity	0	No removal required
Invalid unit price	0	No removal required
Total-value mismatch	20,663 rows on first check	Recalculated from quantity × unit price

After cleaning, the analysis-ready file contained 99,863 rows and all 11 fields passed the final missing-value, duplicate and validity checks. The executed notebook records zero missing values after the promo identifier treatment and zero invalid quantity, price, date or total-value rows.

One important implementation detail is worth keeping visible: the first total-value check did not match because the initial validation formula omitted discount. The cleaning step then used quantity × unit price, which is the formula actually written into the final cleaned dataset. That decision should be understood when interpreting transaction value.

2. What the cleaned data contains

The cleaned sample covers 1 January 2022 to 31 December 2025, with one record per transaction line. It contains 4,998 distinct SKUs and 30 stores. The analysis therefore has enough history to examine both product-level demand and longer seasonal movement.

Field group	Observed scope
Time	2022-01-01 to 2025-12-31 (1,461 days)
SKUs	4,998 in the analysed sales sample
Stores	30
Channels	In-Store, Online, Mobile App
Quantity	1 to 5 per transaction line; no non-positive values
Unit price	■24.41 to ■4,755.47; no non-positive values
Sales value	■117,212,684.34 across the cleaned sample
Transactions	51,863 distinct receipts

The channel mix is also complete: 55,154 lines are In-Store, 30,074 are Online and 14,772 are Mobile App. No channel values were missing. Promotion identifiers were sparse in the raw extract, so missing identifiers were normalised rather than treated as an unknown promotion.

For downstream forecasting, the cleaned transaction data was aggregated into weekly demand at SKU level. The executed notebook produced 80,005 observed week-SKU rows before the later SKU-week grid was completed.

3. Demand pattern over time

The cleaned sample contains 188,034 units of demand and ■117.21 million of sales value. Average daily demand was 128.7 units; average weekly demand was 895.4 units. The highest observed weekly demand was 1,315 units for the week starting 1 December 2025, while the lowest was 207 units in the partial week starting 27 December 2021.

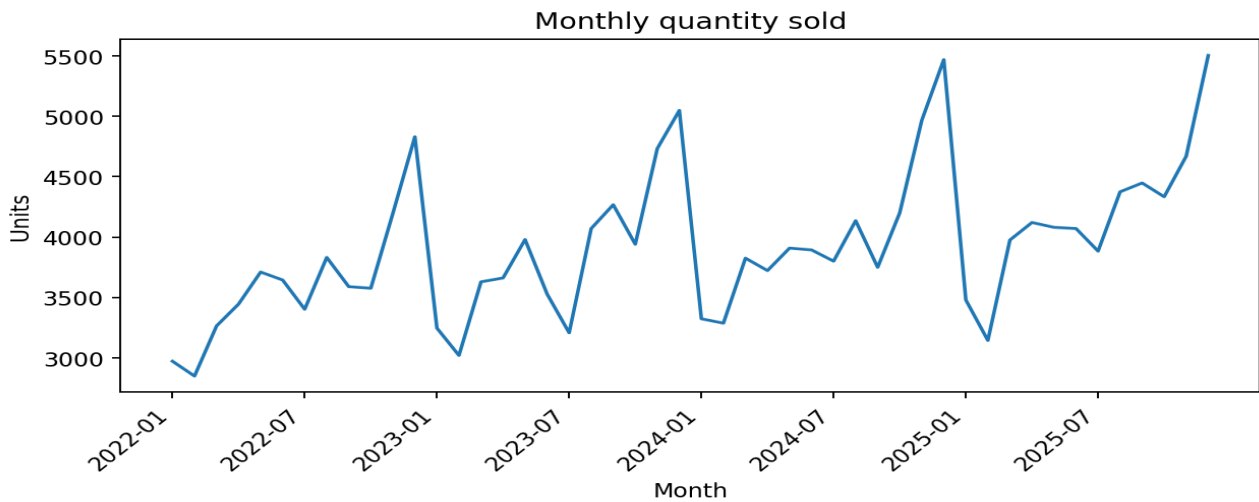


Figure 1. Monthly demand movement across the four-year analysis period.

The monthly series is not flat. The notebook also shows a recurring calendar pattern: December has the highest average monthly quantity at 5,213 units, while February is the lowest at 3,076 units. That recurring difference supports keeping seasonality in the forecasting workflow rather than relying only on a recent moving average.

4. Seasonality and category mix

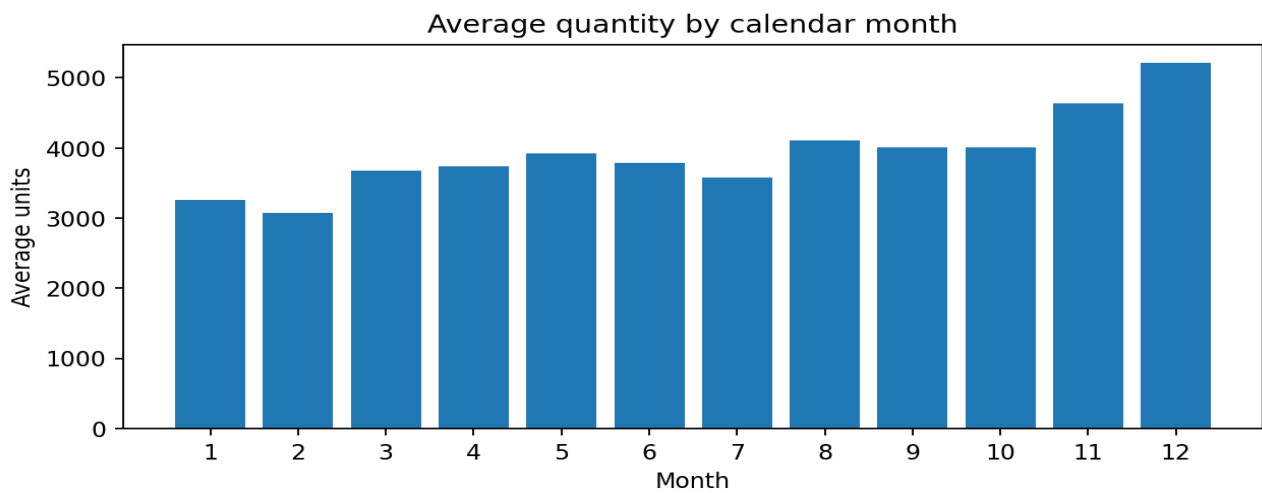


Figure 2. Average quantity by calendar month, calculated across the four years in the cleaned sample.

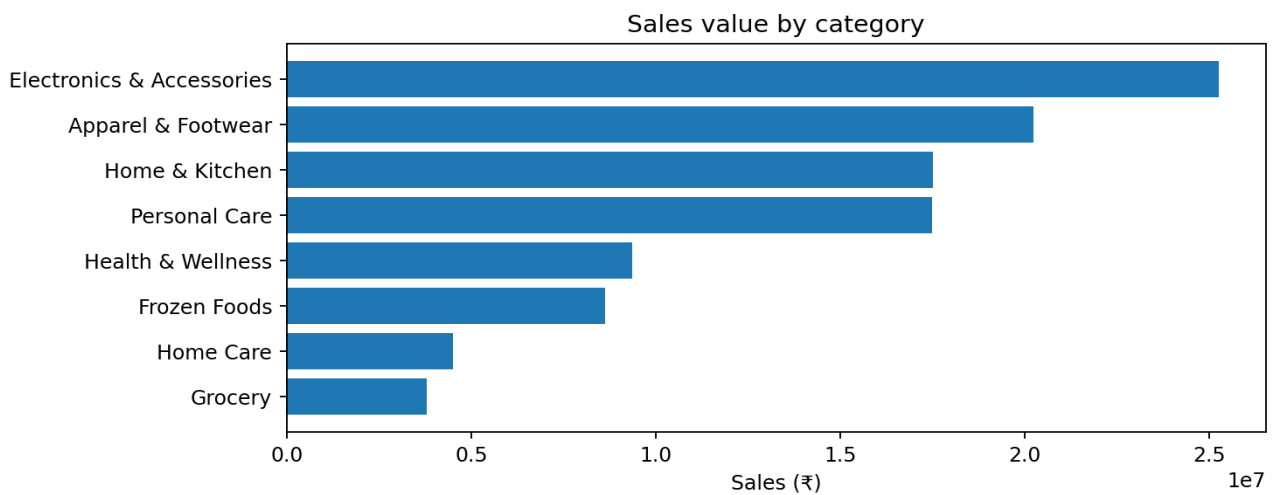


Figure 3. Sales value concentration by category.

Electronics & Accessories generated the largest category sales value at **₹25.26 million**, followed by Apparel & Footwear at **₹20.23 million** and Home & Kitchen at **₹17.52 million**. Personal Care generated the highest quantity at 27,235 units, showing that unit volume and sales value do not tell the same story.

5. Product concentration and slow movers

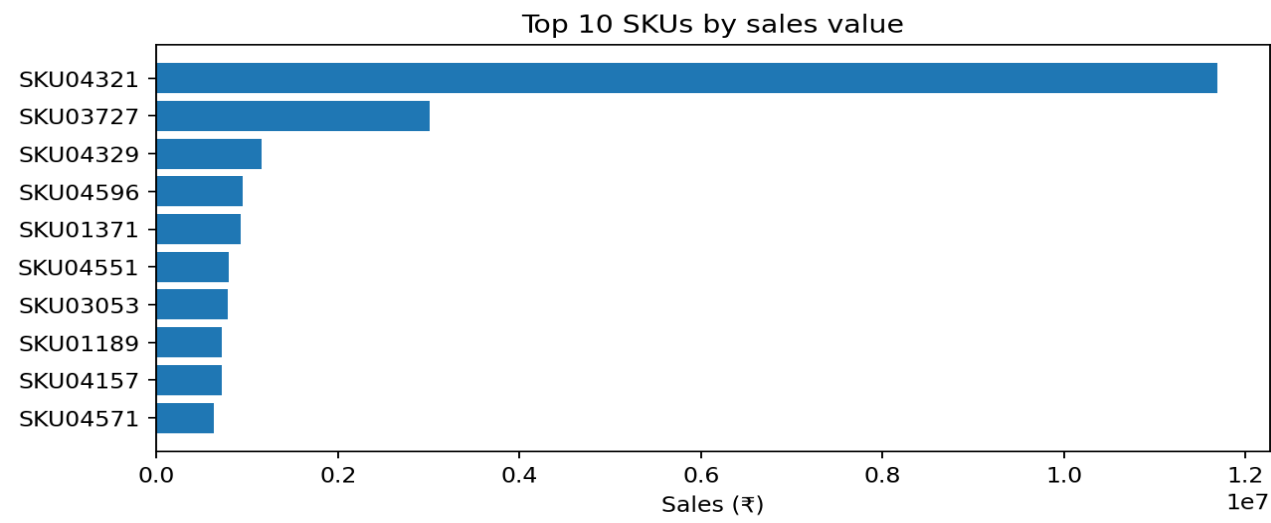


Figure 4. Top 10 SKUs by sales value in the cleaned sample.

SKU04321 is the clearest concentration point in the analysis: 12,156 units and 11.69 million in sales value. Its sales contribution is about 9.97% of the cleaned sample. SKU03727 is next at 3.01 million. The gap between these two products and the rest of the list means availability of a small number of high-value products can have a disproportionate commercial effect.

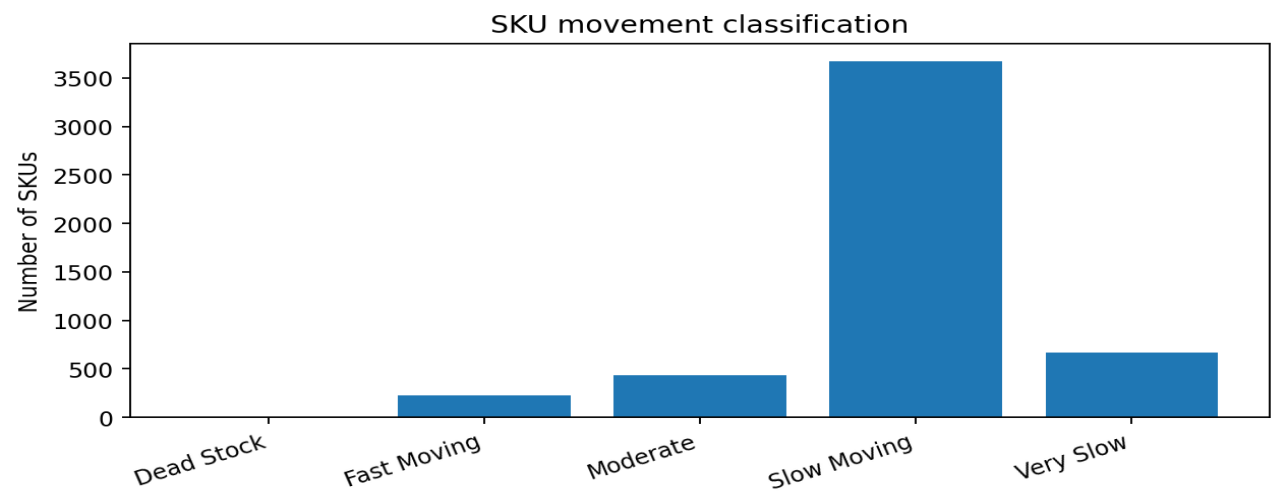


Figure 5. SKU movement classification used in the notebook.

6. Business interpretation and handoff

The EDA gives three practical signals for the next stages of FORESIGHT:

- 1. Protect high-concentration demand.** SKU04321 and the other leading products should be watched closely in the forecast and inventory-risk layers because a stockout on a high-value, high-volume item carries more commercial exposure than a similar unit error on a low-volume SKU.
- 2. Plan for the calendar pattern.** December demand is materially higher than February demand in the four-year sample. The forecasting stage therefore needs a seasonal reference rather than assuming each week behaves like the immediately preceding week.
- 3. Separate slow movement from true dead stock.** The movement analysis classifies 3,668 SKUs as Slow Moving, 669 as Very Slow and 2 as Dead Stock. The two dead-stock SKUs recorded zero sales in the analysed sample. Slow movement should be reviewed with inventory position before a markdown decision is made.

The next stage should use these findings as context rather than as decisions by themselves. The forecasting and risk layers add future demand and inventory position, which is what turns the EDA into an operational recommendation.

Source note: figures in this memo are taken from the executed final EDA notebook and the processed project files available at handoff. The memo intentionally avoids adding claims that were not present in those outputs.